Multi-Variate Analysis

Objective

To analyze inventory datasets and identify patterns, inefficiencies, and insights by using the combinations of three or more columns together.

Overview

Client has been provided 07 csv documents/tables of **“Logistics & Supply Chain – Inventory Management”** that we have to analyze.

In this particular project we have 07 tables to analyze:

Business Domain - Logistics & Supply Chain
Dataset Name - Inventory Management
Business Objectives –

1) Inventory Discrepancy Log
2) Inventory Item Master
3) Inventory KPI Summary
4) Inventory Movement Log
5) Inventory Profiling Dataset
6) Inventory Validation Log
7) Task Allocation Log

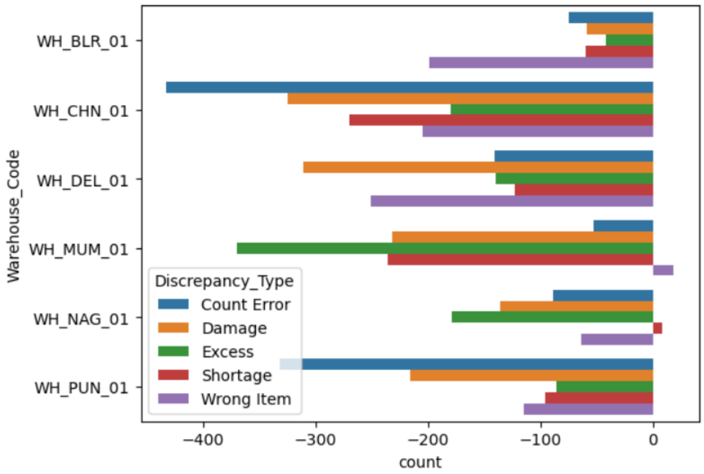
Data Classification Rules:

The following rules are strictly followed as per analytics guidelines:

1. Primary Keys and Foreign Keys are classified as “Identifiers”, not as Categorical data.
2. Identifiers are used only for relationships, joins, and data integrity.
3. Identifiers are excluded from statistical calculations and segmentation logic.
4. Only business-meaningful attributes are classified as Categorical, Numerical, Geographical, or Date-Time.
5. The given dataset has records between Jan 2021 to Dec 2024.

1. Table - inventory_discrepancy_log_large

➢ Warehouse Wise Variance Quantity by different Discrepancy types:



This Bar plot clearly shows that,
“Count Error” discrepancies are more at Chennai & Pune Warehouse,
Mumbai warehouse has higher “Excess” discrepancies,
Chennai, Delhi, Mumbai& Pune Warehouse have more “Damage” discrepancies
But Mumbai Warehouse have some “Wrong Item” whereas Nagpur Warehouse have extra stock in reserve that means no shortage.
Below are the detailed counts of discrepancies within various warehouses.

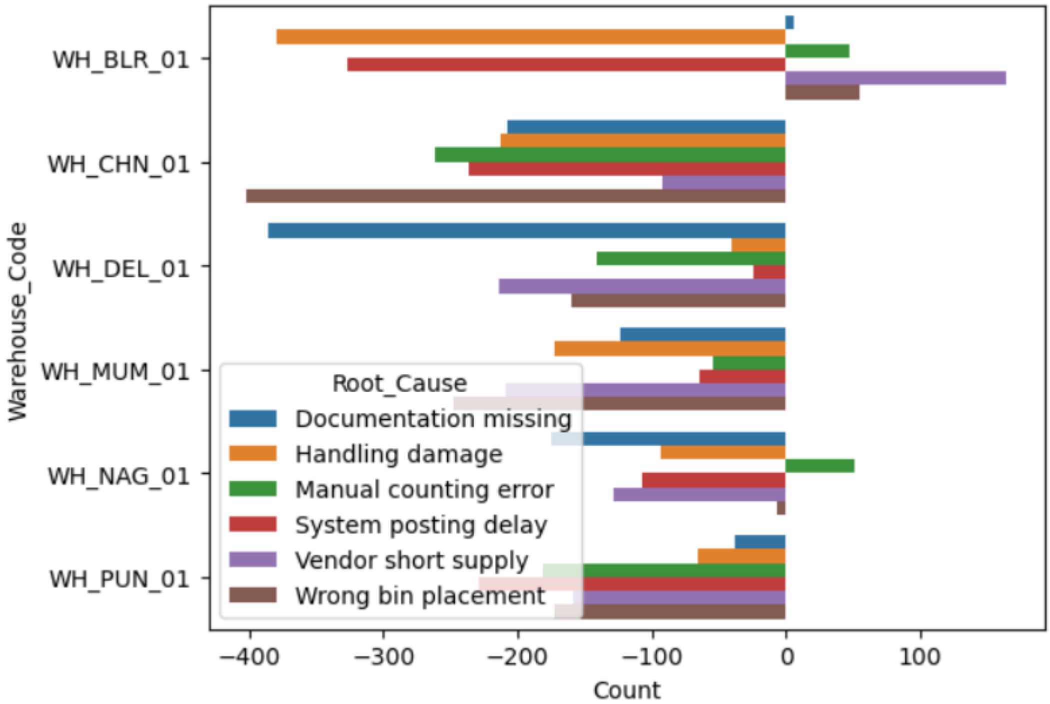
Warehouse_Code	Discrepancy_Type	count
WH_BLR_01	Count Error	-75
WH_CHN_01	Count Error	-433
WH_DEL_01	Count Error	-141
WH_MUM_01	Count Error	-53
WH_NAG_01	Count Error	-89
WH_PUN_01	Count Error	-332

Warehouse_Code	Discrepancy_Type	count
WH_BLR_01	Excess	-42
WH_CHN_01	Excess	-180
WH_DEL_01	Excess	-140
WH_MUM_01	Excess	-370
WH_NAG_01	Excess	-179
WH_PUN_01	Excess	-86

Warehouse_Code	Discrepancy_Type	count
WH_BLR_01	Damage	-59
WH_CHN_01	Damage	-325
WH_DEL_01	Damage	-311
WH_MUM_01	Damage	-232
WH_NAG_01	Damage	-136
WH_PUN_01	Damage	-216

Warehouse_Code	Discrepancy_Type	count
WH_BLR_01	Shortage	-60
WH_CHN_01	Shortage	-270
WH_DEL_01	Shortage	-123
WH_MUM_01	Shortage	-236
WH_NAG_01	Shortage	8
WH_PUN_01	Shortage	-96

➢ Warehouse Wise Variance Quantity by different Root Cause:



This Bar plot shows that,
“Handling Damage” root cause are more at Bangalore Warehouse,
Chennai warehouse has higher “Wrong Bin Placement” root cause,
Delhi Warehouse have more “Documentation Missing” root cause
But Bangalore & Nagpur Warehouse have some “Manual Counting Error” whereas Bangalore Warehouse have extra stock in reserve under “Vendor Short Supply” & “Wrong Bin Placement” root cause.
Below are the detailed counts of root cause of discrepancies within various warehouses.

Warehouse_Code	Root_Cause	Count
WH_BLR_01	Wrong bin placement	55
WH_CHN_01	Wrong bin placement	-402
WH_DEL_01	Wrong bin placement	-160
WH_MUM_01	Wrong bin placement	-248
WH_NAG_01	Wrong bin placement	-7
WH_PUN_01	Wrong bin placement	-173

Warehouse_Code	Root_Cause	Count
WH_BLR_01	Documentation missing	6
WH_CHN_01	Documentation missing	-208
WH_DEL_01	Documentation missing	-386
WH_MUM_01	Documentation missing	-124
WH_NAG_01	Documentation missing	-175
WH_PUN_01	Documentation missing	-38

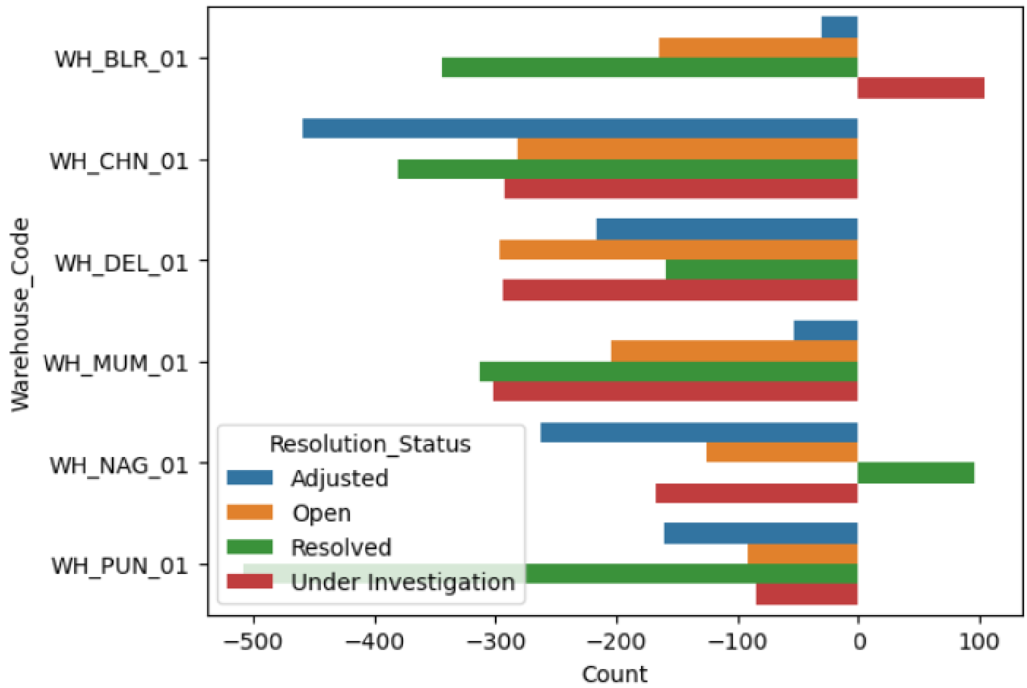
Warehouse_Code	Root_Cause	Count
WH_BLR_01	Handling damage	-380
WH_CHN_01	Handling damage	-212
WH_DEL_01	Handling damage	-41
WH_MUM_01	Handling damage	-173
WH_NAG_01	Handling damage	-93
WH_PUN_01	Handling damage	-66

Warehouse_Code	Root_Cause	Count
WH_BLR_01	Vendor short supply	164
WH_CHN_01	Vendor short supply	-92
WH_DEL_01	Vendor short supply	-214
WH_MUM_01	Vendor short supply	-209
WH_NAG_01	Vendor short supply	-129
WH_PUN_01	Vendor short supply	-158

Warehouse_Code	Root_Cause	Count
WH_BLR_01	System posting delay	-327
WH_CHN_01	System posting delay	-237
WH_DEL_01	System posting delay	-24
WH_MUM_01	System posting delay	-64
WH_NAG_01	System posting delay	-107
WH_PUN_01	System posting delay	-229

Warehouse_Code	Root_Cause	Count
WH_BLR_01	Manual counting error	47
WH_CHN_01	Manual counting error	-262
WH_DEL_01	Manual counting error	-141
WH_MUM_01	Manual counting error	-55
WH_NAG_01	Manual counting error	51
WH_PUN_01	Manual counting error	-181

➢ Warehouse Wise Variance Quantity by Resolution Status:



This Bar plot shows that,
“Adjusted” discrepancies are more at Chennai Warehouse,
“Resolved” discrepancies are more at Pune location.
Chennai warehouse has higher “Wrong Bin Placement” root cause,
Nagpur warehouse has some positive “Resolved” discrepancies recorded &
Bangalore warehouse has some positive “Under Investigation” discrepancies.
Below are the detailed counts of Resolution Status of discrepancies within various warehouses.

Warehouse_Code	Resolution_Status	Count
WH_BLR_01	Under Investigation	104
WH_CHN_01	Under Investigation	-292
WH_DEL_01	Under Investigation	-294
WH_MUM_01	Under Investigation	-302
WH_NAG_01	Under Investigation	-167
WH_PUN_01	Under Investigation	-85

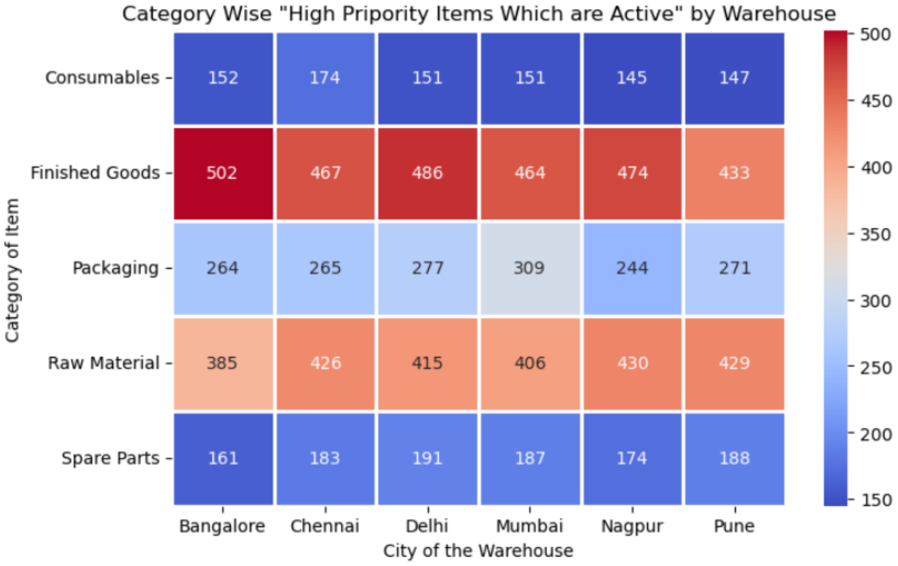
Warehouse_Code	Resolution_Status	Count
WH_BLR_01	Resolved	-344
WH_CHN_01	Resolved	-380
WH_DEL_01	Resolved	-159
WH_MUM_01	Resolved	-313
WH_NAG_01	Resolved	96
WH_PUN_01	Resolved	-508

Warehouse_Code	Resolution_Status	Count
WH_BLR_01	Open	-165
WH_CHN_01	Open	-282
WH_DEL_01	Open	-297
WH_MUM_01	Open	-204
WH_NAG_01	Open	-126
WH_PUN_01	Open	-91

Warehouse_Code	Resolution_Status	Count
WH_BLR_01	Adjusted	-30
WH_CHN_01	Adjusted	-459
WH_DEL_01	Adjusted	-216
WH_MUM_01	Adjusted	-54
WH_NAG_01	Adjusted	-263
WH_PUN_01	Adjusted	-161

2. Table - inventory_item_master_large

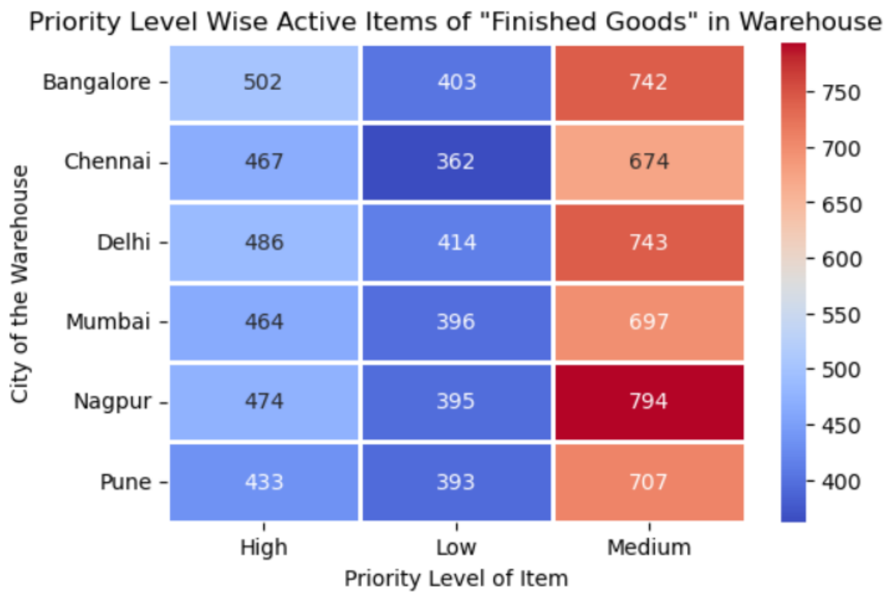
➢ **How much Items are available with High Priority & Active flag by various Warehouses:**



Above heatmap is created on the basic of **High Priority** Item & **Active** flag items only.

By watching out this chart we clarify that, in all warehouse's Finished goods are mostly stored whereas Consumables and Spare Parts are stored in minimal quantity.

Bangalore warehouse stores a higher unit of Finished Goods (i.e. 502) And Pune stores less.



Above heatmap is created on the basic of **Finished Goods** & **Active** flag items only.

By watching out this chart we clarify that, in all warehouse's Medium Priority level items are stored more and Low priority level items are stored less.

Nagpur Warehouse have more medium priority level items (i.e. 794) & Chennai Warehouse have less low priority level items (i.e. 362).

That means most of the Finished Goods in inventory are medium level priority.

3. Table - inventory_kpi_summary_large

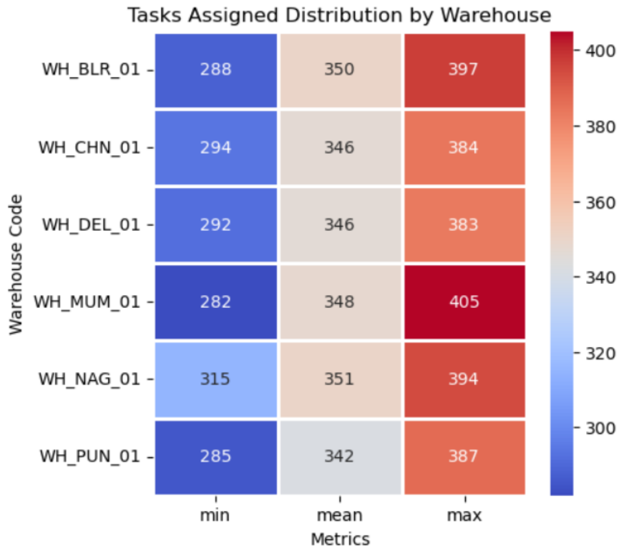


Fig - A

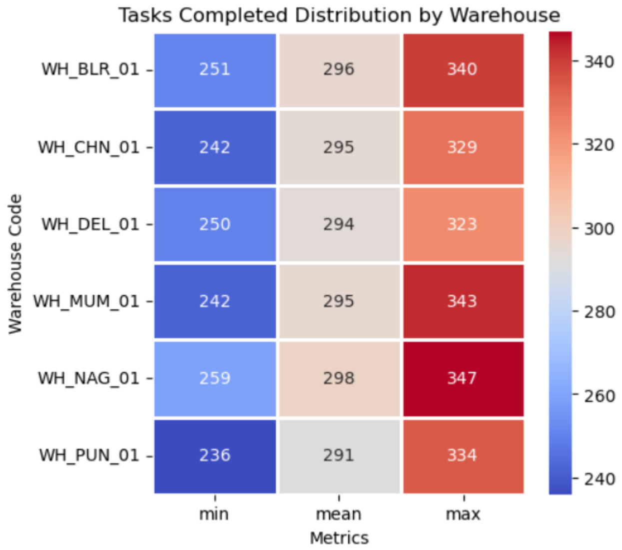


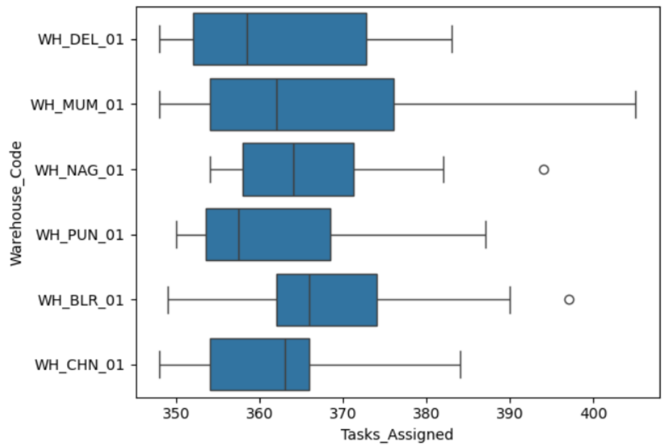
Fig - B

Above heatmaps shows the Task assigned & Task completion distribution on the basis of minimum, maximum & average task. So, Mumbai warehouse has been assigned more tasks (i.e. 405) as compared to others whereas Mumbai & Nagpur has maximum task completion ratio as compared to other warehouses.

And we found that, the **average** task assigned tasks combining to all warehouses is **347** and the **average** task completion count is **294**.
That's why we decided to analyse only above average tasks, so we sort the data whose task assigned above average (i.e 347 task) and task completed above average (i.e 294 task).

Below boxplot shows total counts of tasks assigned above average by respective warehouse.

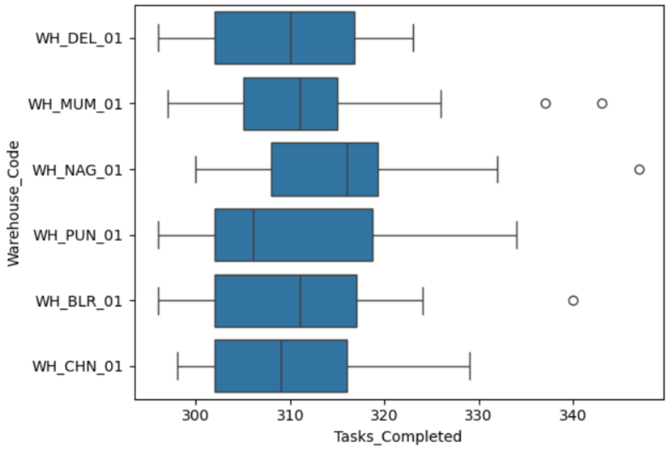
Warehouse_Code	count
WH_BLR_01	9205
WH_CHN_01	7608
WH_DEL_01	7980
WH_MUM_01	7688
WH_NAG_01	8788
WH_PUN_01	6517



By watching above boxplot, we noticed most of assigned task is lies between 355 to 375 tasks. Count figure clearly shows that Bangalore warehouse has been assigned more quantity of tasks (i.e. 9205 tasks) and Pune warehouse has been assigned less quantity of task (i.e. 6517 tasks). So, we conclude that, Bangalore has more tasks to perform as compared to other warehouses.

Below boxplot shows that, the total counts of tasks completed above average by respective warehouse.

Warehouse_Code	count
WH_BLR_01	7772
WH_CHN_01	6508
WH_DEL_01	6814
WH_MUM_01	6560
WH_NAG_01	7566
WH_PUN_01	5592



By watching above boxplot, we noticed most of completed task is lies between 302 to 320 tasks. Count figure clearly shows that Bangalore warehouse has been completed more quantity of tasks (i.e. 7772 tasks) and Pune warehouse has been completed less quantity of task (i.e. 5592 tasks). So, we conclude that, Bangalore warehouse is overpressured as compared to other warehouses.

- Average Task completion rate is 84 to 85 % across all the warehouses.

```
Warehouse_Code
WH_BLR_01     84.476042
WH_CHN_01     85.042083
WH_DEL_01     84.825208
WH_MUM_01     84.884583
WH_NAG_01     85.002708
WH_PUN_01     85.106458
Name: Task_Completion_Rate_%,
```

- Average Mismatch rate is 9 to 10 % across all the warehouses.

```
Warehouse_Code
WH_BLR_01     10.294792
WH_CHN_01     10.368333
WH_DEL_01     10.007083
WH_MUM_01      9.788542
WH_NAG_01     10.226667
WH_PUN_01      9.843750
Name: Mismatch_Rate_%,
```

- Average Validation Failure rate 21.88 % across all the warehouses.

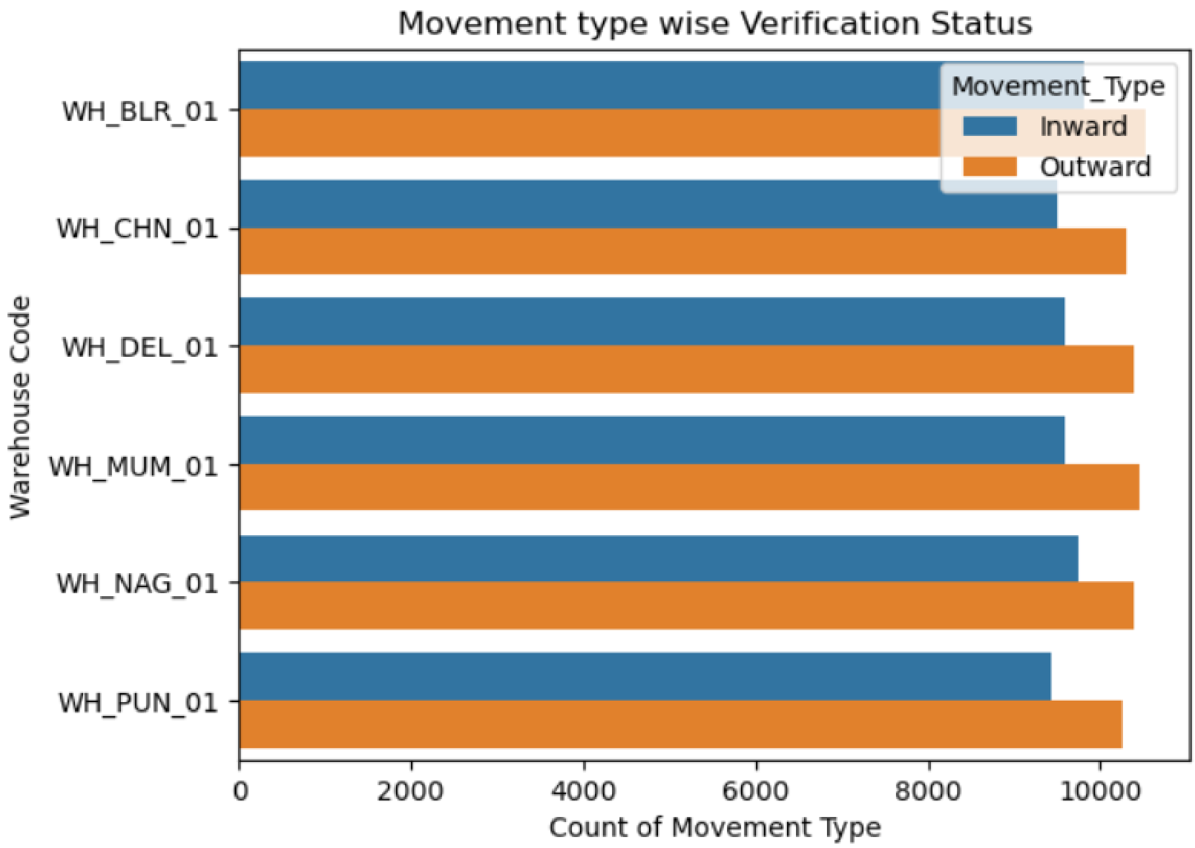
```
Warehouse_Code
WH_BLR_01     21.885417
WH_CHN_01     21.885417
WH_DEL_01     21.885417
WH_MUM_01     21.885417
WH_NAG_01     21.885417
WH_PUN_01     21.885417
Name: Validation_Failure_Rate_%,
```

4. Table - inventory_movement_log_large

Warehouse_Code	Movement_Type	
WH_BLR_01	Inward	19.530371
	Outward	19.450917
WH_CHN_01	Inward	19.801875
	Outward	19.686031
WH_DEL_01	Inward	19.638683
	Outward	19.675423
WH_MUM_01	Inward	19.780290
	Outward	19.604037
WH_NAG_01	Inward	19.246561
	Outward	19.478830
WH_PUN_01	Inward	19.837964
	Outward	19.625730

This figure clearly concludes the average movement quantity for both Inward and Outward movement is around 19.

Below figure shows the count of movement type (Inward/Outward) with respect to particular warehouses.



5. Table - inventory_profiling_large

This sheet provides the data of inventory profiling between 2023 & 2024 near equally distributed, and by this we can analyse the behaviour of items present in the inventory.
By provided sheet, we clearly know that most of the Finished Goods are stored in the storage Zone-A (i.e. 4412 items) and less items are stored in Zone- C and overflow storage zone.

Now, we are going to analyse this profiling data on the basis of **category** of the item and its **priority level**, mostly we analyse the high priority level items in the inventory.

The detailed count of item category wise is as follows:

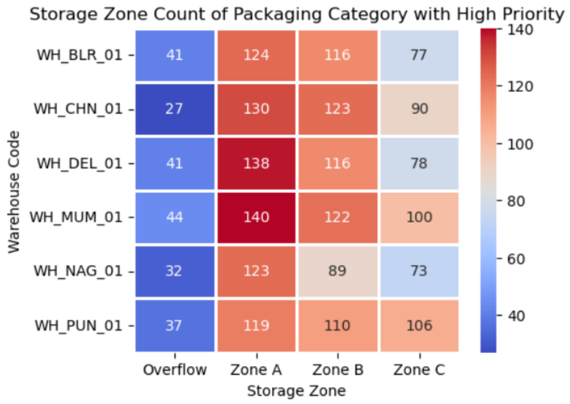


Fig – (A)

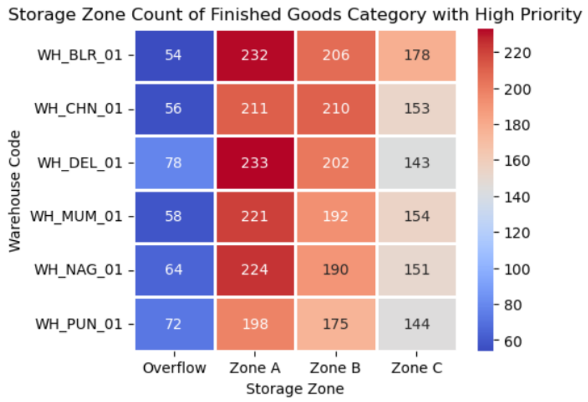


Fig – (B)

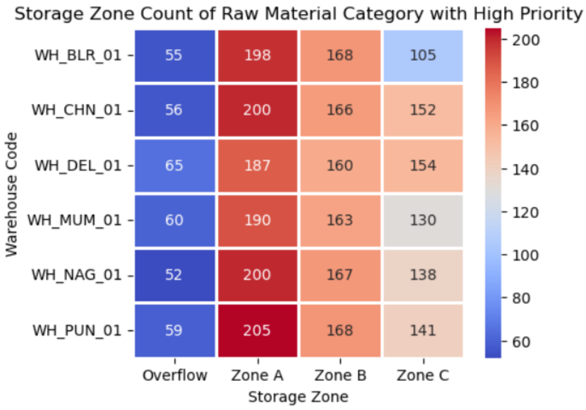


Fig – (C)

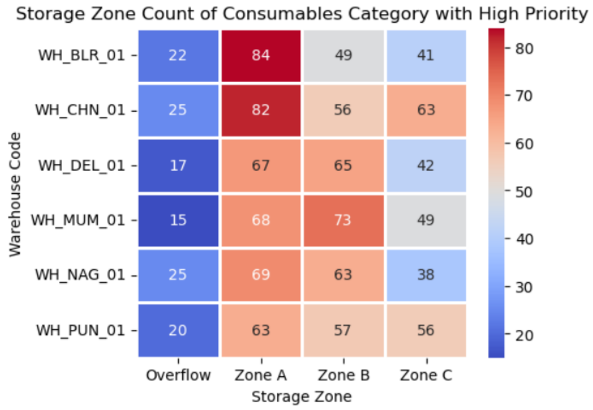


Fig – (D)

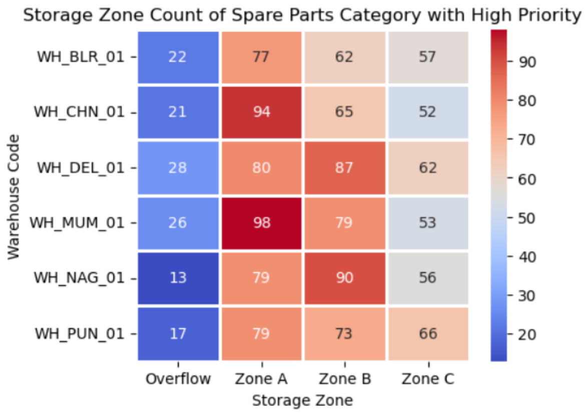


Fig – (E)

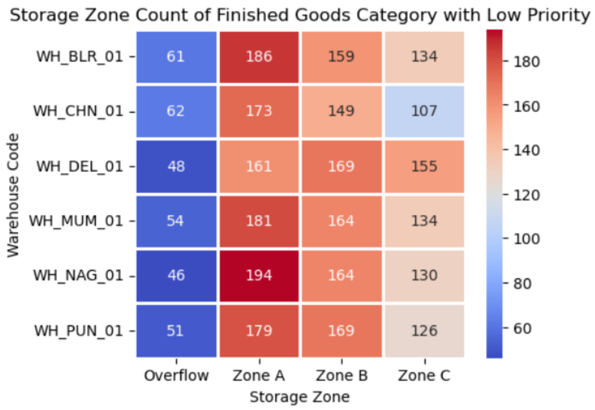


Fig – (F)

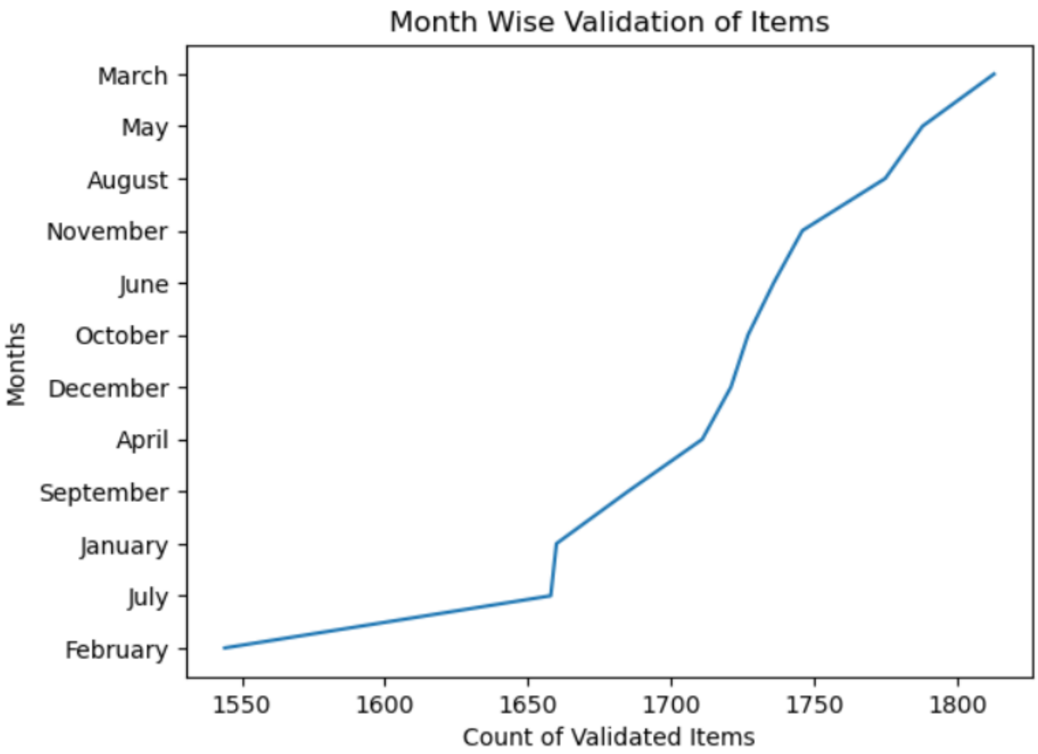
Above mentioned heatmap concludes that,

- Most of the inventory items are stored in Storage Zone A & B.
- Fig-(A) clarifies Mumbai & Delhi warehouse stores most of **packaging** items in Zone - A which have **higher** priority level (i.e. 140 & 138 items).
- Fig-(F) clarifies Nagpur warehouse stores most of **Finished Good** items in Zone - A which have **lower** priority level (i.e. 194 items).
- Fig-(B) clarifies Bangalore & Delhi warehouse stores most of **Finished Good** items in Zone - A which have **higher** priority level (i.e. 233 & 232 items).
- Fig-(C) clarifies Pune warehouse stores most of **Raw Material** items in Zone - A which have **higher** priority level (i.e. 205 items).
- Fig-(D) clarifies Bangalore & Chennai warehouse stores most of **Consumable** items in Zone - A which have **higher** priority level (i.e. 84 & 84 items).
- Fig-(E) clarifies Mumbai & Chennai warehouse stores most of **Spare part** items in Zone - A which have **higher** priority level (i.e. 98 & 94 items).

6. Table - inventory_validation_log_large

This sheet provides the data of inventory movement validation between 2021 & 2024 slightly by increasing numbers year by year, and by this we can ensures data accuracy through validation checks.

By provided sheet, we clearly observed that, most of validation is done in the month of **"March"** (i.e. 1813) and in the month of **"February",** lower validation was done (i.e. 1544). The detailed validation by month is as follows:



Now we are going to analyze the distribution of validated items by year and validator based on the selected Validation Rule and Validation Status.

- ❖ **Distribution of validated items by Year and Validator based on the "UOM consistency Validation Rule" & Validation Status :-**

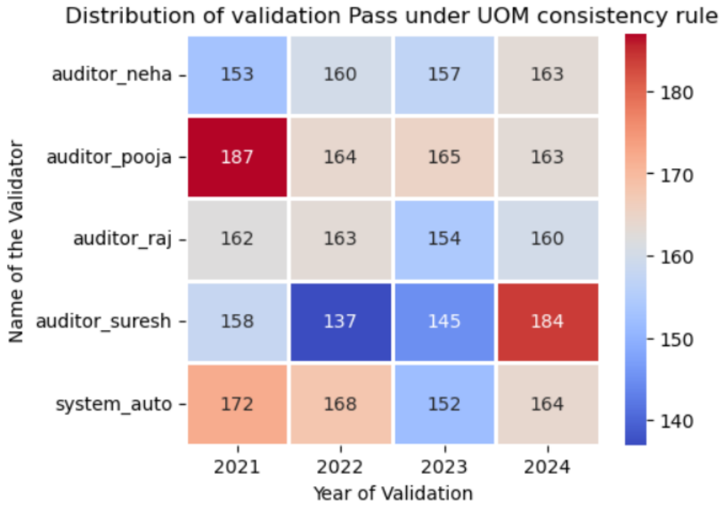


Fig- (A) Validation Status = Pass

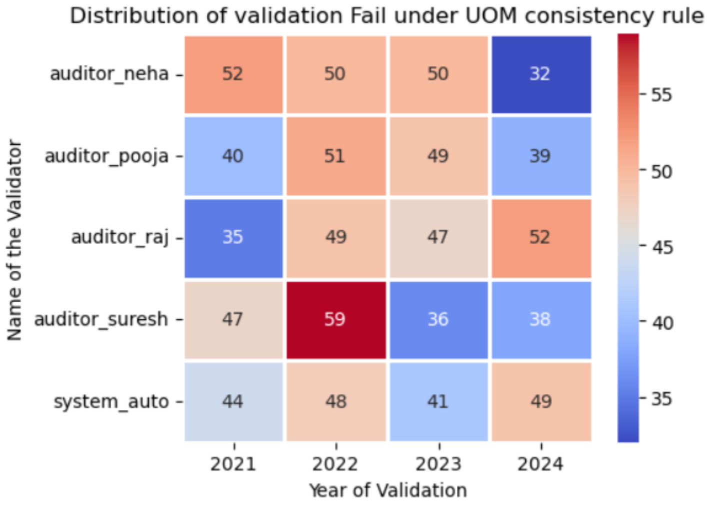


Fig- (B) Validation Status = Fail

- ✓ Figure (A) illustrates the validator-wise and year-wise movement validation count that **passed** under the "UOM Consistency" validation rule.

  In 2021 & 2024, Auditor Pooja and Auditor Suresh recorded the highest number of **passed** movement validations, with 187 and 184 validations, respectively. In 2022, Auditor Suresh recorded the lowest number of **passed** movement validations, with 137 validations under the "UOM Consistency" validation rule.

- ✓ Figure (B) illustrates the validator-wise and year-wise movement validation count that **failed** under the "UOM Consistency" validation rule.

  In 2022, Auditor Suresh recorded the highest number of **failed** movement validations, with 59 validations. In 2024, Auditor Neha recorded the lowest number of **failed** movement validations, with 32 validations under the "UOM Consistency" validation rule.

❖ Distribution of validated items by Year and Validator based on the "ERP Posting validation Rule" & Validation Status :-

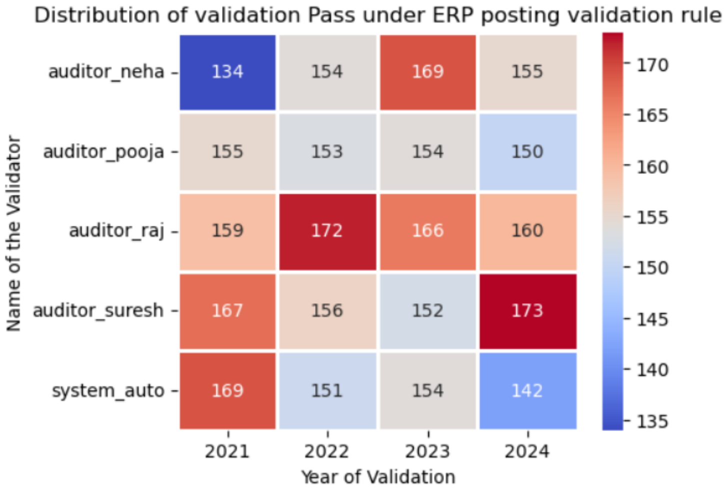


Fig- (A) Validation Status = Pass

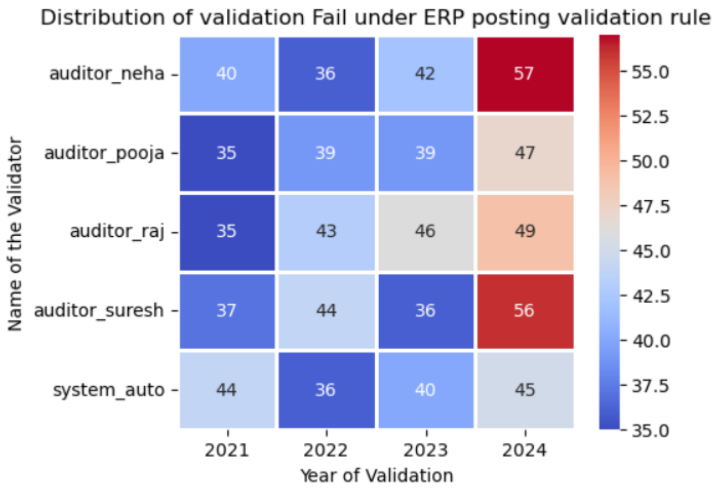


Fig- (B) Validation Status = Fail

- ✓ Figure (A) illustrates the validator-wise and year-wise movement validation count that **passed** under the "ERP Posting validation Rule".

  In 2022 & 2024, Auditor Raj and Auditor Suresh recorded the highest number of **passed** movement validations, with 172 and 173 validations, respectively. In 2021, Auditor Neha recorded the lowest number of **passed** movement validations, with 134 validations under the "ERP Posting validation Rule".

- ✓ Figure (B) illustrates the validator-wise and year-wise movement validation count that **failed** under the "ERP Posting validation Rule".

  In 2024, Auditor Suresh and Auditor Neha recorded the highest number of failed movement validations, with 56 and 57 validations, respectively. In 2021, Auditor Pooja & Raj recorded the lowest number of **failed** movement validations, with 35 validations under the "ERP Posting validation Rule".

❖ Distribution of validated items by Year and Validator based on the "Document completeness validation Rule" & Validation Status :-

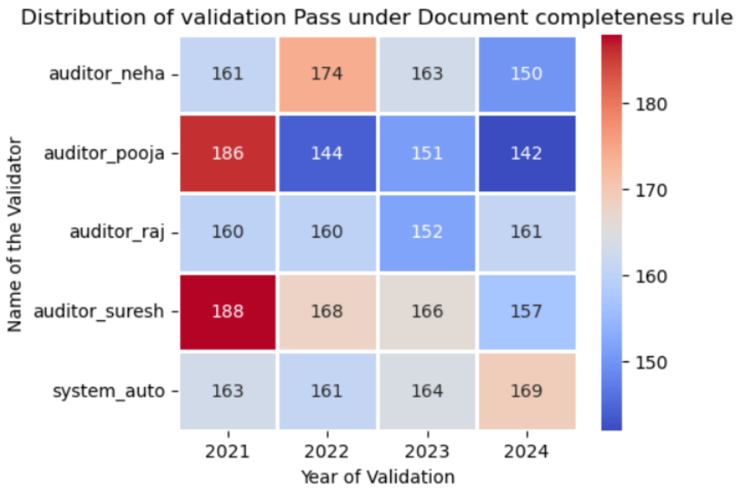


Fig- (A) Validation Status = Pass

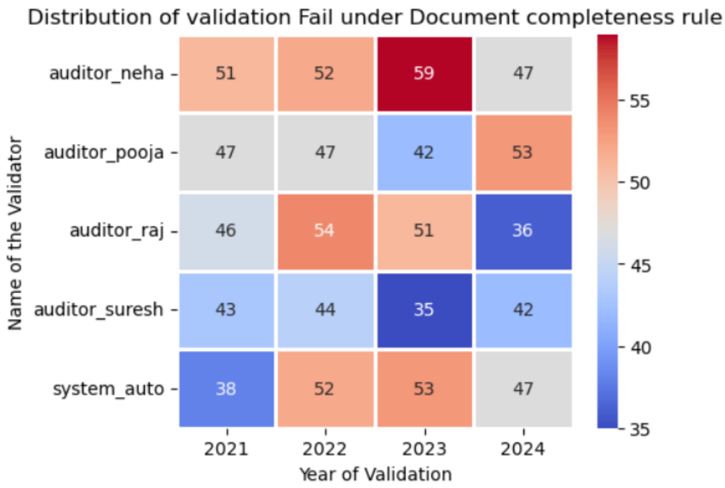


Fig- (B) Validation Status = Fail

- ✓ Figure (A) illustrates the validator-wise and year-wise movement validation count that **passed** under the "Document completeness validation Rule".

  In 2021, Auditor Suresh and Auditor Pooja recorded the highest number of **passed** movement validations, with 188 and 186 validations, respectively. In 2024, Auditor Pooja recorded the lowest number of **passed** movement validations, with 142 validations under the "Document completeness validation Rule".

- ✓ Figure (B) illustrates the validator-wise and year-wise movement validation count that **failed** under the "Document completeness validation Rule".

  In 2023, Auditor Neha recorded the highest number of failed movement validations, with 59 validations. In 2023 & 2024, Auditor Suresh and Auditor Raj recorded the highest number of failed movement validations, with 35 and 36 validations, respectively under the "Document completeness validation Rule".

❖ Distribution of validated items by Year and Validator based on the "Warehouse-Location match validation Rule" & Validation Status :-

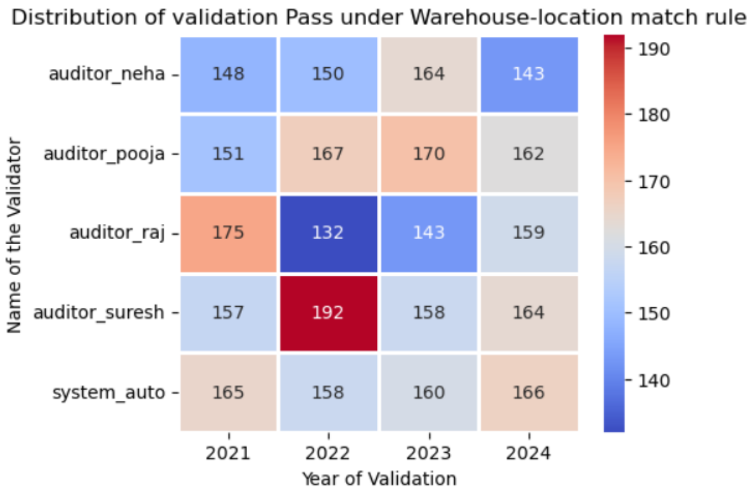


Fig- (A) Validation Status = Pass

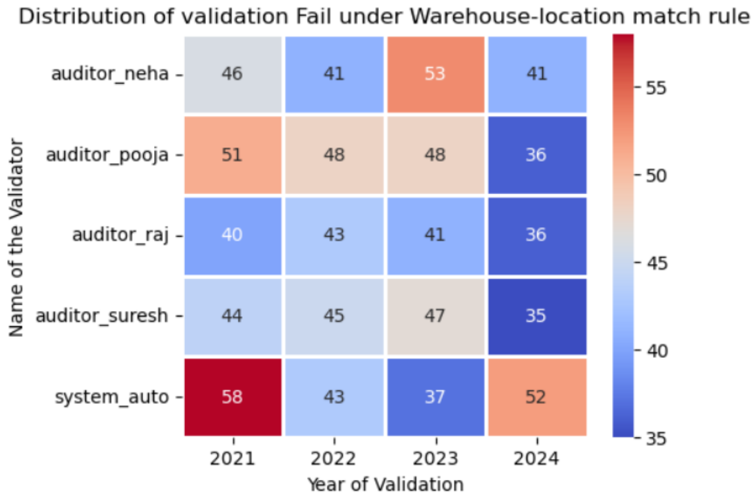


Fig- (B) Validation Status = Fail

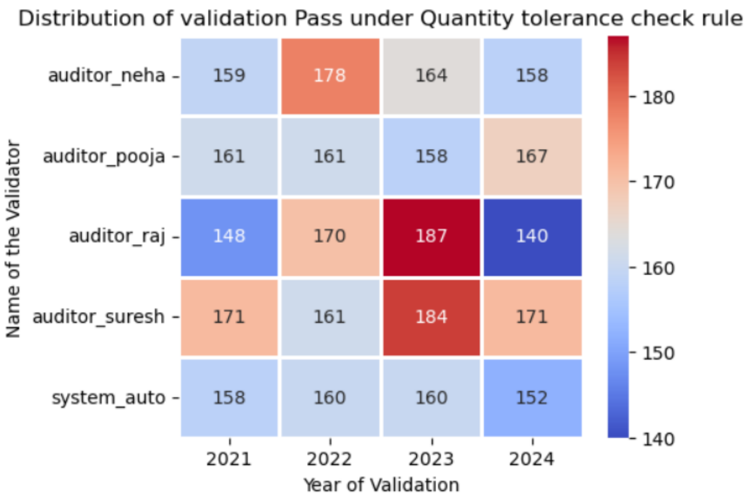
- ✓ Figure (A) illustrates the validator-wise and year-wise movement validation count that **passed** under the "Warehouse-Location match validation Rule".

  In 2022, Auditor Suresh recorded the highest number of **passed** movement validations, with 192 validations. In 2022, Auditor Raj recorded the lowest number of **passed** movement validations, with 132 validations under the "Warehouse-Location match validation Rule".

- ✓ Figure (B) illustrates the validator-wise and year-wise movement validation count that **failed** under the "Warehouse-Location match validation Rule".

  In 2021, System Auto recorded the highest number of failed movement validations, with 58 validations. In 2024, Auditor Suresh, Raj and Pooja recorded the highest number of failed movement validations, with around 36 validations under the "Warehouse-Location match validation Rule".

❖ Distribution of validated items by Year and Validator based on the "Quantity Tolerance Check validation Rule" & Validation Status :-



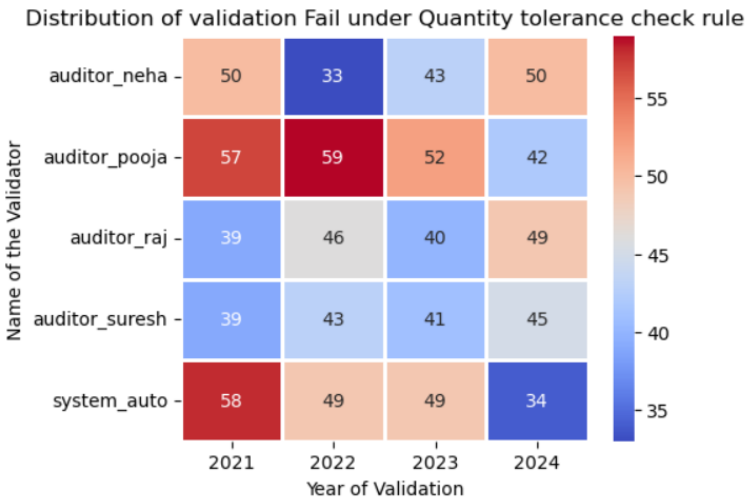


Fig- (A) Validation Status = Pass

Fig- (B) Validation Status = Fail

- ✓ Figure (A) illustrates the validator-wise and year-wise movement validation count that **passed** under the "Quantity Tolerance Check validation Rule".

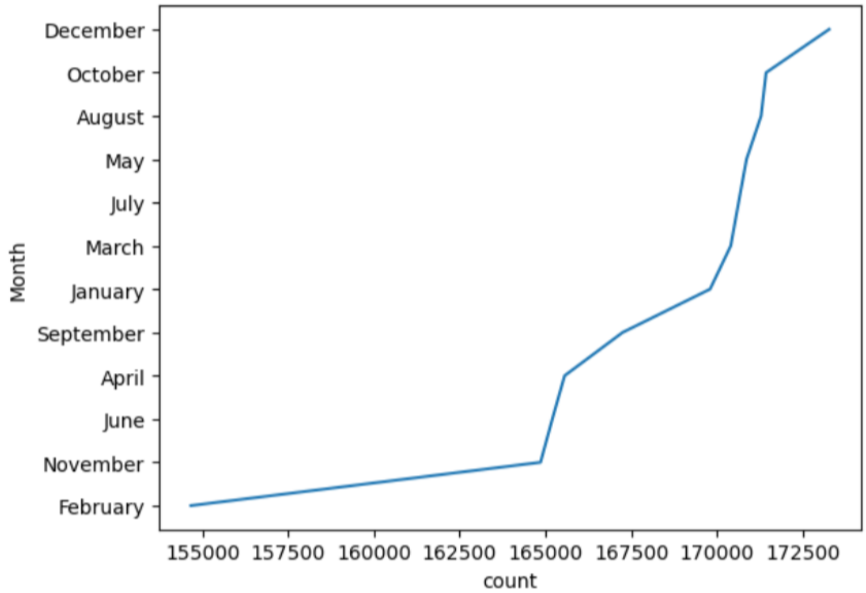
  In 2023, Auditor Suresh and Auditor Raj recorded the highest number of passed movement validations, with 184 and 187 validations, respectively. In 2024, Auditor Raj recorded the lowest number of passed movement validations, with 140 validations under the "Quantity Tolerance Check validation Rule".

- ✓ Figure (B) illustrates the validator-wise and year-wise movement validation count that **failed** under the "Quantity Tolerance Check validation Rule".

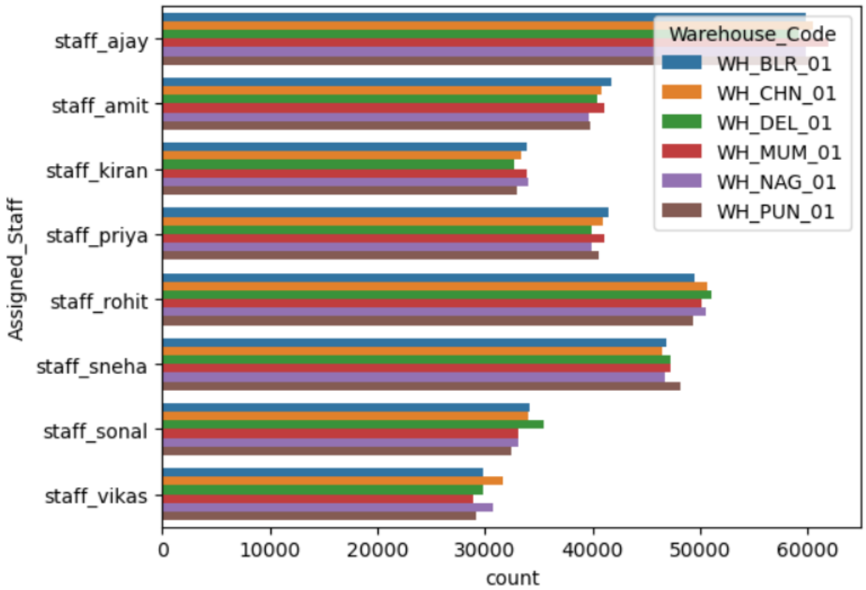
  In 2021 & 2022, System Auto & Auditor Pooja recorded the highest number of failed movement validations, with around 57 to 58 validations. In 2023 & 2024, Auditor Neha and System Auto recorded the highest number of failed movement validations, with 33 and 34 validations, respectively under the "Quantity Tolerance Check validation Rule".

7. Table - task_allocation_log_large

- This sheet provides the data of task assignment and workforce performance between the year 2021 & 2024.
  By provided sheet, we clearly observed that, Quantity wise most of task allocation is done in the month of **"December"** (i.e. 1,73,264 tasks) and in the month of **"February",** lower task allocation was done (i.e. 1,54,670 tasks).
  The detailed allocation by month is as follows:



- Below bar plot clearly declares that, Staff Ajay has been assigned higher tasks quantity (i.e. 3,61,343 tasks) whereas to Staff Vikas assigned lower quantity of tasks (i.e. 1,79,949 tasks).
  Average assigned quantity per task to all staff members are 19 to 20 tasks.



➢ Staff wise Analysis of Task Allocation:

Now we are going to analyse the task allocation and performance of assigned 08 staffs across all the warehouses and all task types:

❖ Staff – Ajay

Staff – Ajay has been assigned a total of **17,944** tasks and total quantity of **3,61,343** tasks.

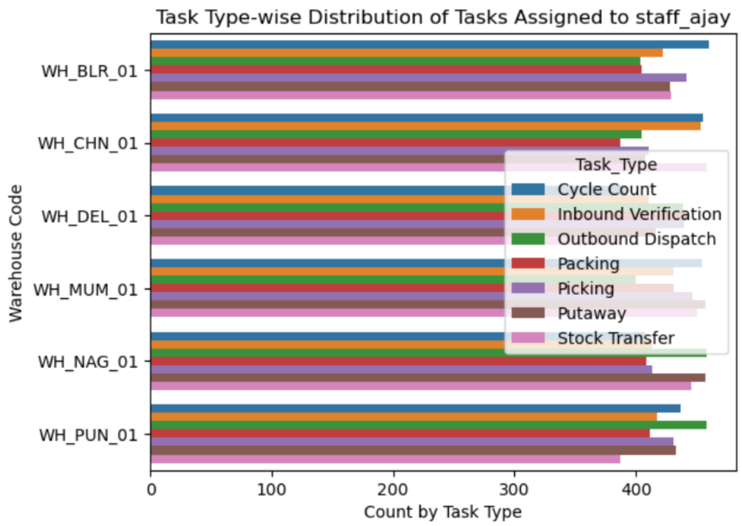


Fig-(A)

The Fig-(A) bar chart shows that tasks assigned to staff_ajay are fairly evenly distributed across all warehouses and task types, indicating an equally balanced workload. However, certain tasks such as Picking, Putaway, and Stock Transfer have slightly higher counts in warehouses like WH_NAG_01 and WH_CHN_01. Overall, staff_ajay handles a consistent volume of tasks across locations, with no major imbalance in task allocation.

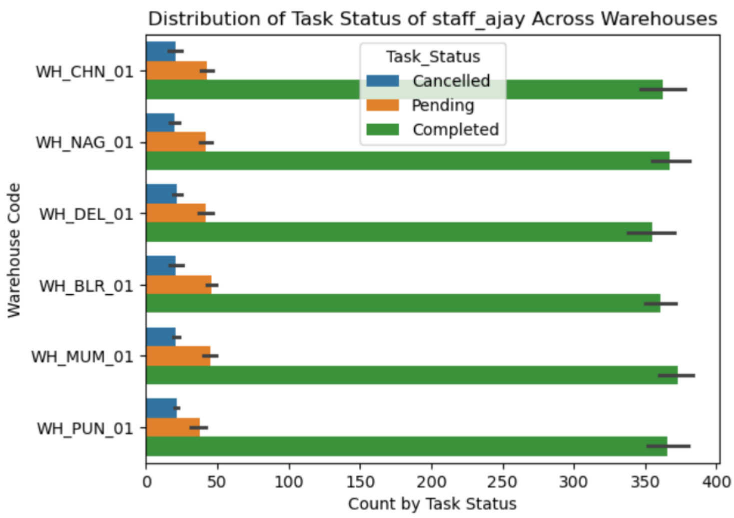


Fig-(B)

The Fig-(B) chart shows that for staff_ajay, the majority of tasks across all warehouses are marked as Completed, while Pending and Cancelled tasks are significantly lower. This indicates strong task completion performance and efficient workflow management. Slightly higher completed tasks can be observed in warehouses like WH_MUM_01 and WH_NAG_01, but overall, the task status distribution remains consistent across all locations.

❖ Staff – Sonal

Staff – Ajay has been assigned a total of **9,946** tasks and total quantity of **2,02,219** tasks.

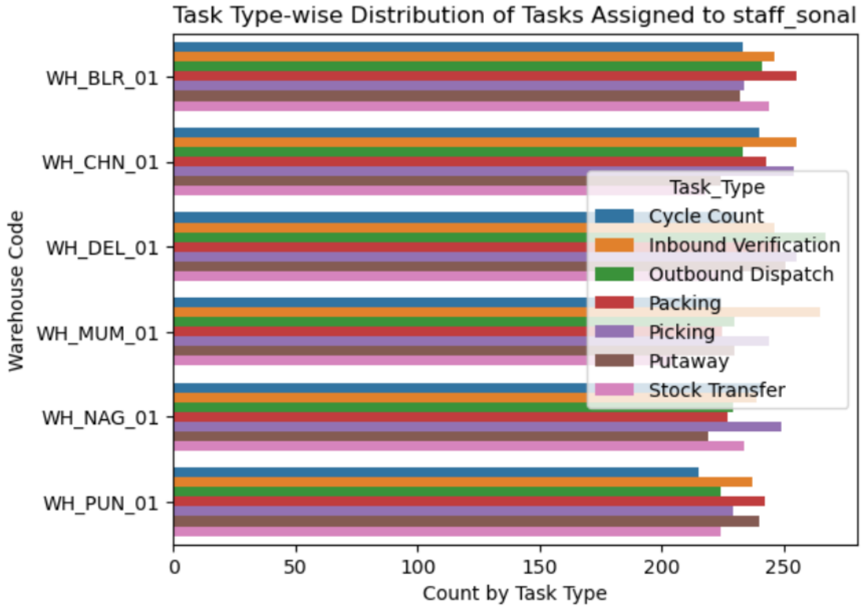


Fig-(A)

The Fig-(A) bar chart shows that tasks assigned to staff_sonal are fairly evenly distributed across all warehouses and task types, indicating an equally balanced workload. However, certain tasks such as Packing, Inbound Dispatch, and picking have slightly higher counts in warehouses like WH_DEL_01 and WH_MUM_01. Overall, staff_sonal handles a consistent volume of tasks assigned across locations, with no major imbalance in task allocation.

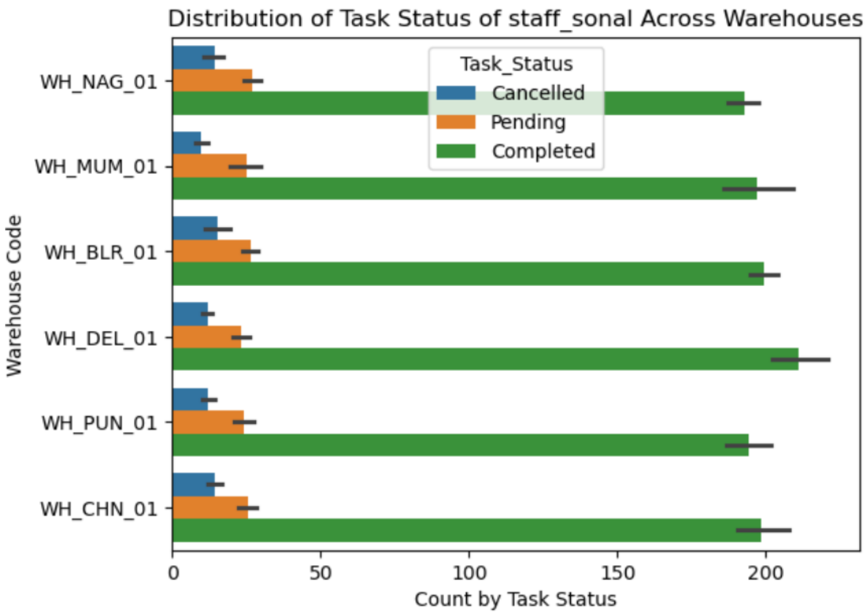


Fig-(B)

The Fig-(B) chart shows that for staff_sonal, the majority of tasks across all warehouses are marked as Completed, while Pending and Cancelled tasks are significantly lower. This indicates strong task completion performance and efficient workflow management. Slightly higher completed tasks can be observed in warehouses like WH_DEL_01 and WH_CHN_01, but overall, the task status distribution remains consistent across all locations.

❖ Staff – Vikas

Staff – Ajay has been assigned a total of **8,952** tasks and total quantity of **1,79,949** tasks.

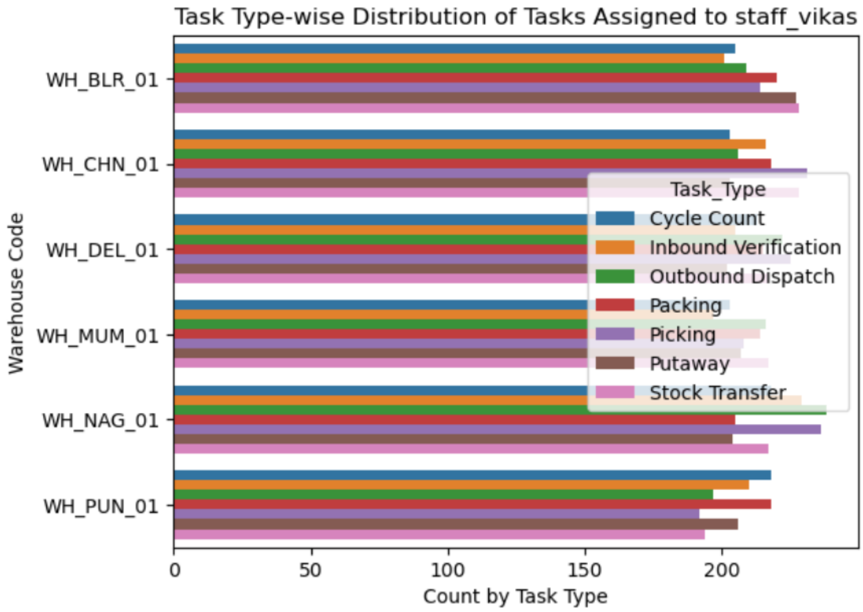


Fig-(A)

The Fig-(A) bar chart shows that tasks assigned to staff_vikas are fairly evenly distributed across all warehouses and task types, indicating an equally balanced workload. However, certain tasks such as Picking, outbound Dispatch, and packing have slightly higher counts in warehouses like WH_NAG_01 and WH_CHN_01. Overall, staff_vikas handles a consistent volume of tasks assigned across locations, with no major imbalance in task allocation.

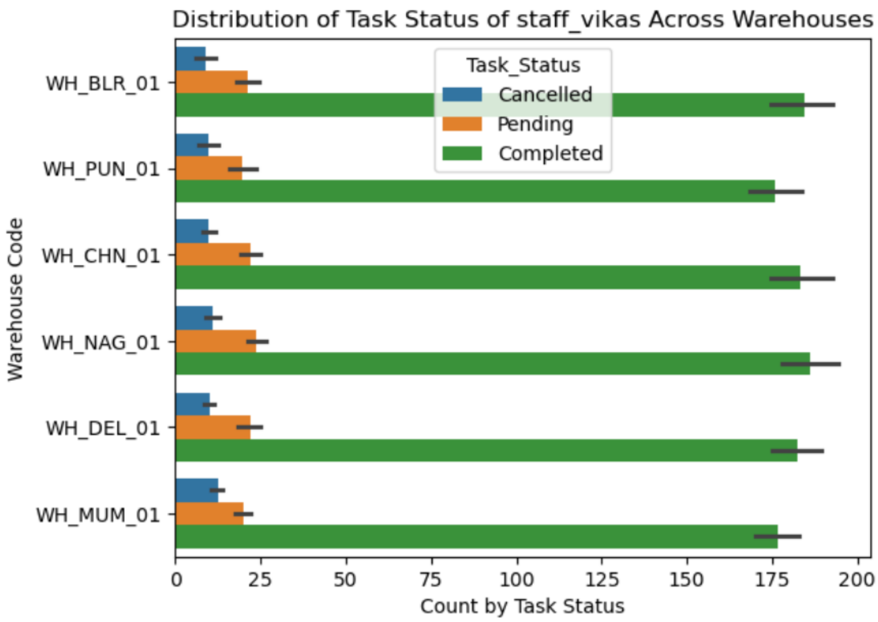


Fig-(B)

The Fig-(B) chart shows that for staff_vikas, the majority of tasks across all warehouses are marked as Completed, while Pending and Cancelled tasks are significantly lower. This indicates strong task completion performance and efficient workflow management. Slightly higher completed tasks can be observed in warehouses like WH_NAG_01 and WH_BLR_01, but overall, the task status distribution remains consistent across all locations.

❖ Staff – Sneha

Staff – Ajay has been assigned a total of **14,004** tasks and total quantity of **2,82,569** tasks.

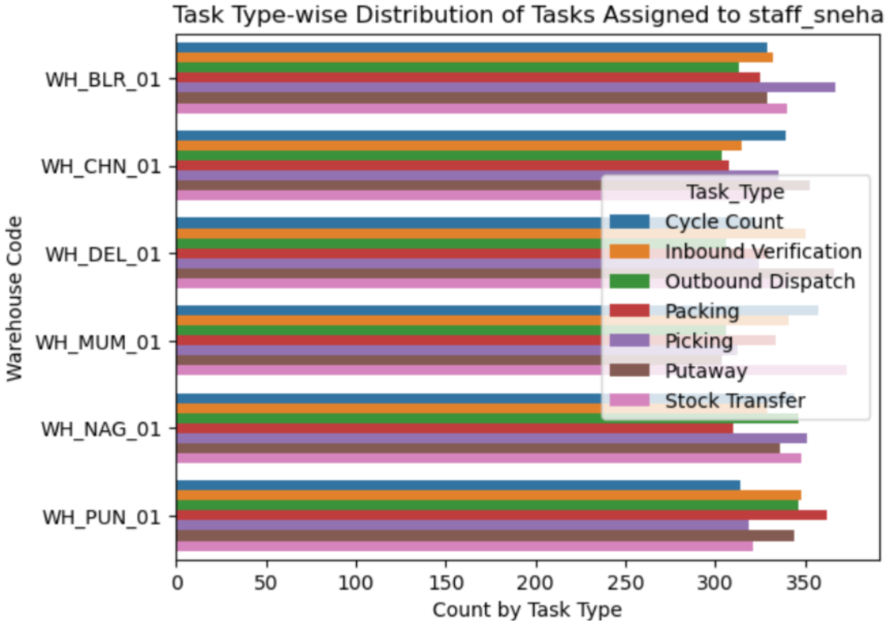


Fig-(A)

The Fig-(A) bar chart shows that tasks assigned to staff_sneha are fairly evenly distributed across all warehouses and task types, indicating an equally balanced workload. However, certain tasks such as Picking, stock transfer, and packing have slightly higher counts in warehouses like WH_BLR_01 and WH_MUM_01. Overall, staff_sneha handles a consistent volume of tasks assigned across locations, with no major imbalance in task allocation.

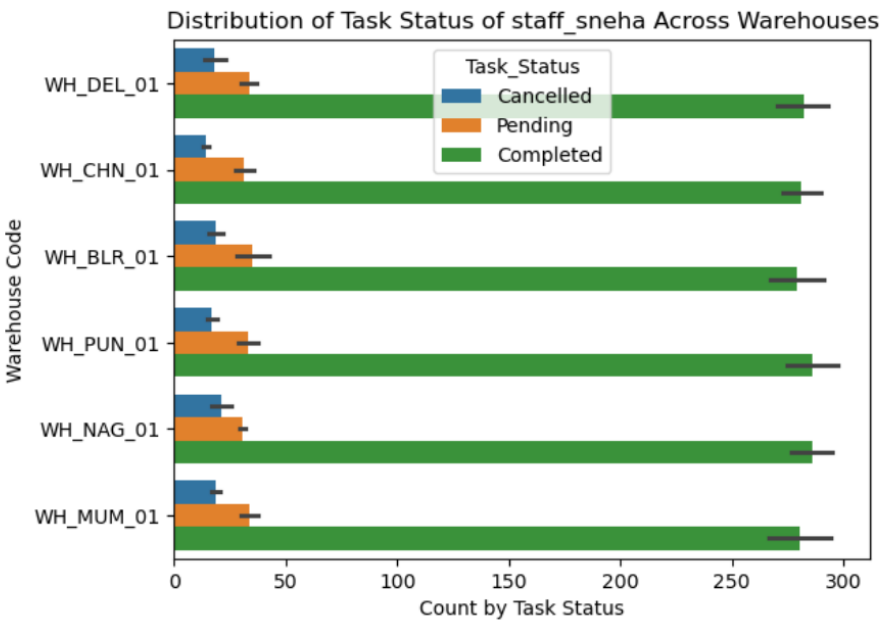


Fig-(B)

The Fig-(B) chart shows that for staff_sneha, the majority of tasks across all warehouses are marked as Completed, while Pending and Cancelled tasks are significantly lower. This indicates strong task completion performance and efficient workflow management. Slightly higher completed tasks can be observed in warehouses like WH_PUN_01 and WH_NAG_01, but overall, the task status distribution remains consistent across all locations.

❖ Staff – Kiran

Staff – Ajay has been assigned a total of **9,991** tasks and total quantity of **2,00,756** tasks.

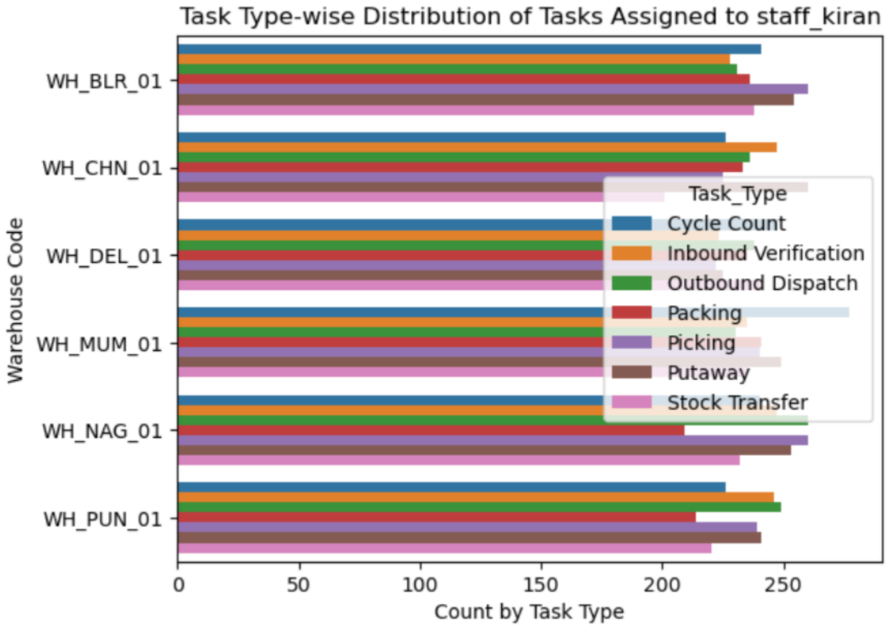


Fig-(A)

The Fig-(A) bar chart shows that tasks assigned to staff_kiran are fairly evenly distributed across all warehouses and task types, indicating an equally balanced workload. However, certain tasks such as Cycle count, putaway, and picking have slightly higher counts in warehouses like WH_MUM_01 and WH_BLR_01. Overall, staff_kiran handles a consistent volume of tasks assigned across locations, with no major imbalance in task allocation.

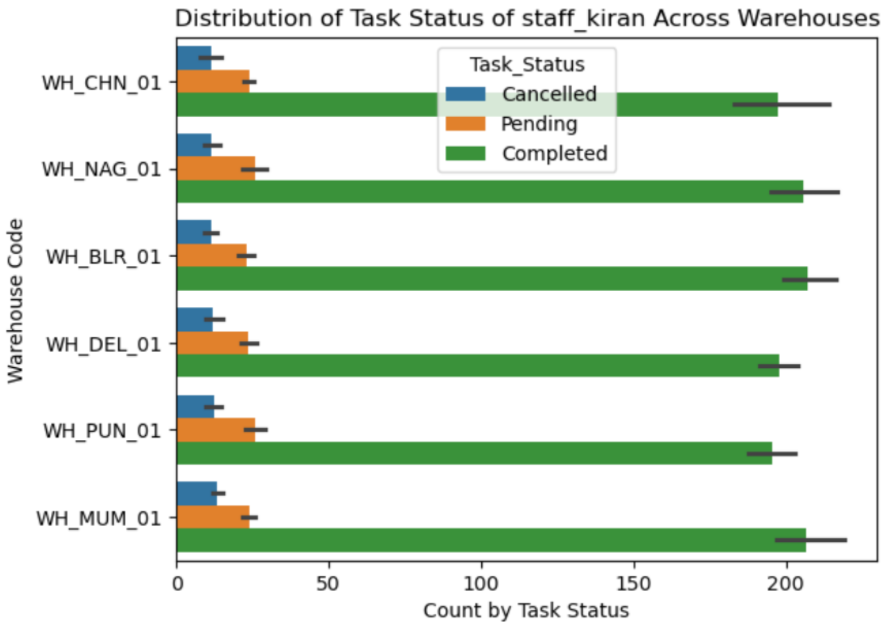


Fig-(B)

The Fig-(B) chart shows that for staff_kiran, the majority of tasks across all warehouses are marked as Completed, while Pending and Cancelled tasks are significantly lower. This indicates strong task completion performance and efficient workflow management. Slightly higher completed tasks can be observed in warehouses like WH_MUM_01 and WH_BLR_01, but overall, the task status distribution remains consistent across all locations.

❖ Staff – Rohit

Staff – Ajay has been assigned a total of **15,050** tasks and total quantity of **3,01,013** tasks.

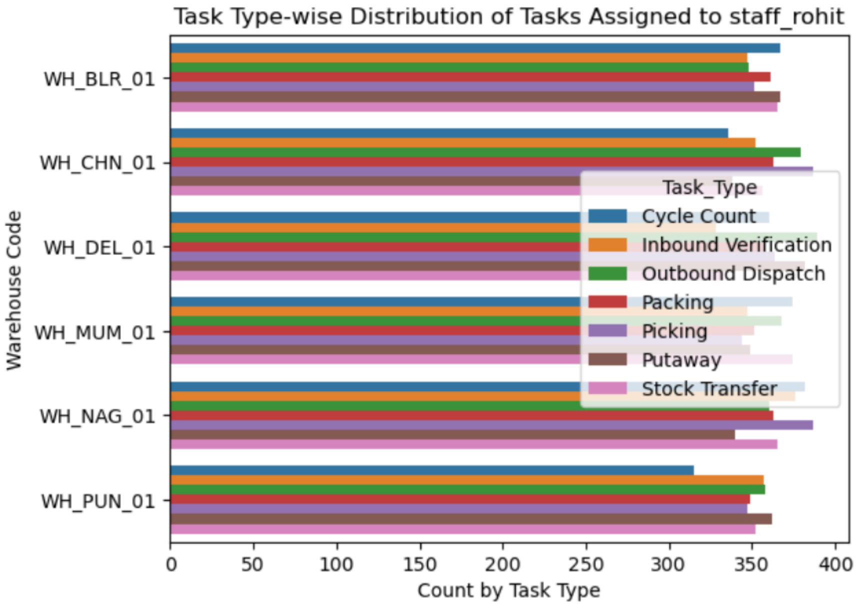


Fig-(A)

The Fig-(A) bar chart shows that tasks assigned to staff_rohit are fairly evenly distributed across all warehouses and task types, indicating an equally balanced workload. However, certain tasks such as Outbound Dispatch, cycle count, and picking have slightly higher counts in warehouses like WH_DEL_01 and WH_NAG_01. Overall, staff_rohit handles a consistent volume of tasks assigned across locations, with no major imbalance in task allocation.

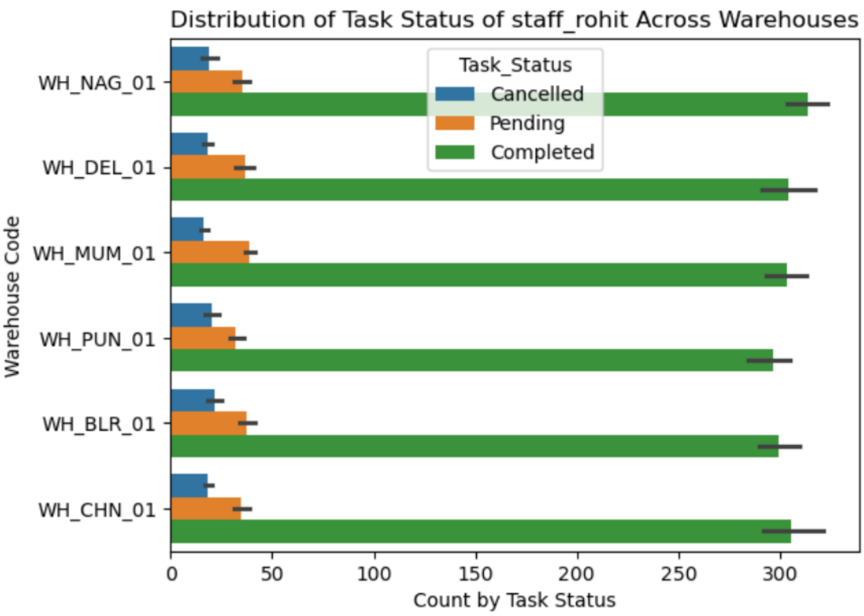


Fig-(B)

The Fig-(B) chart shows that for staff_rohit, the majority of tasks across all warehouses are marked as Completed, while Pending and Cancelled tasks are significantly lower. This indicates strong task completion performance and efficient workflow management. Slightly higher completed tasks can be observed in warehouses like WH_NAG_01 and WH_CHN_01, but overall, the task status distribution remains consistent across all locations.

❖ Staff – Priya

Staff – Ajay has been assigned a total of **12,107** tasks and total quantity of **2,43,836** tasks.

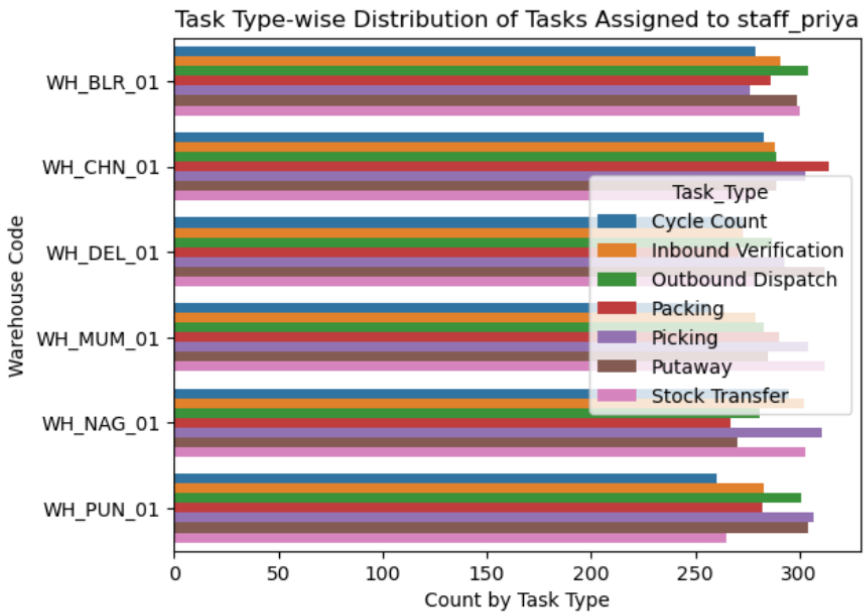


Fig-(A)

The Fig-(A) bar chart shows that tasks assigned to staff_priya are fairly evenly distributed across all warehouses and task types, indicating an equally balanced workload. However, certain tasks such as Picking, Putaway, and packing have slightly higher counts in warehouses like WH_CHN_01 and WH_DEL_01. Overall, staff_priya handles a consistent volume of tasks assigned across locations, with no major imbalance in task allocation.

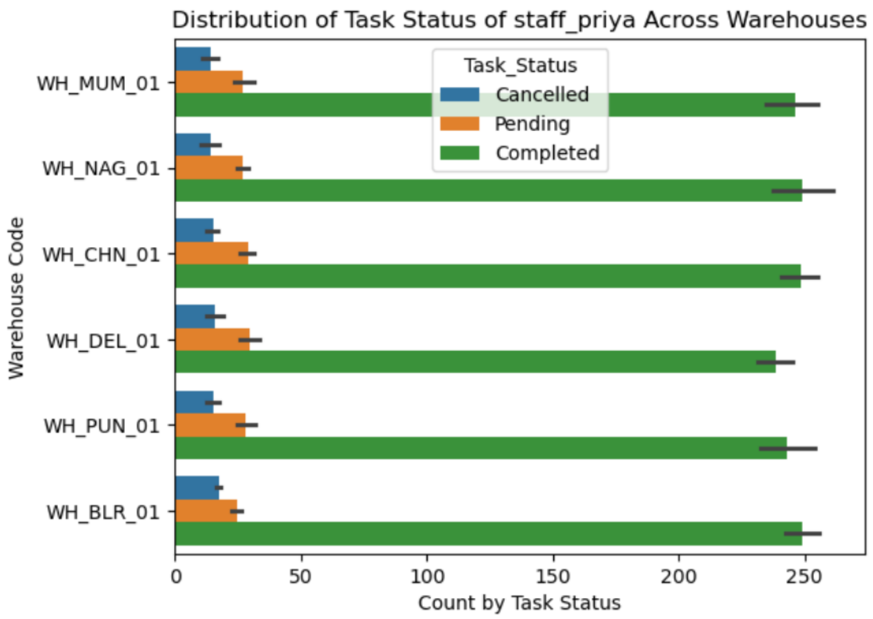


Fig-(B)

The Fig-(B) chart shows that for staff_priya, the majority of tasks across all warehouses are marked as Completed, while Pending and Cancelled tasks are significantly lower. This indicates strong task completion performance and efficient workflow management. Slightly higher completed tasks can be observed in warehouses like WH_NAG_01 and WH_BLR_01, but overall, the task status distribution remains consistent across all locations.